

HSUAN-YU LIU

📍 Taipei, Taiwan | ✉️ hsuanyuu.liu@gmail.com | in sherryliuyu | 🌐 ryukyu coding

Education

National Taiwan University Taipei, Taiwan
B.B.A. in Information Management (cGPA: 4.18/4.3) Expected June 2027

- **Honors:** Academic Achievement Award (×2, ranked 1st/55 in Fall 2025); President's Award (2025)
- **Leadership:** General Coordinator of IM Night (2025), Head of Publicity at IM Student Assoc. (2024-2025)
- **Relevant Coursework:** Data Structure and Algorithm, Database, Web Programming (1st/150), Linux, Machine Learning, Artificial Intelligence, Computer Vision with Deep Learning, Cloud Native

Work & Research Experience

Mitacs Globalink Research Intern, Montréal HCI Group, Université de Montréal June 2026 – Present

- Selected for the Mitacs Globalink Research Internship (~CAD 12,000 fully-funded, competitive international program) to conduct HCI research under Prof. Damien Masson
- Designing a graphical, AI-assisted music composition interface enabling non-expert users to directly manipulate melody, harmony, and rhythm without requiring music production skills

Software Engineer Intern, Printage, Inc. Feb 2026 – Present

- Shipped 9 production features for Floze, an AI companion app with 250K+ users and 20K DAU, using Vue.js, TypeScript, and AWS Lambda (serverless) across MySQL (Prisma), DynamoDB, and Redis
- Built a Meta Ads batch automation tool cutting campaign creation time from ~4 hours to under 10 minutes, replacing manual per-campaign duplication with one-time batch setup
- Integrated AI-driven ad performance analysis, auto-generating daily reports and optimization recommendations to improve marketing ROI

Research Assistant, WIDM Lab, National Central University July 2025 – Present

- Built a natural language-to-n8n agentic pipeline: GPT-4o intent detection, fine-tuned LLM for node operations (Create/Modify/Delete/Insert), and ChromaDB-backed RAG over a self-constructed Knowledge Graph
- Achieved 14.3% / 47.7% gains in Node / Connection F1 over pure-LLM baseline; paper in preparation

Project Experience

Full-Stack Developer, Tangyuan — NTU Study Abroad & Community Platform Dec 2025 – Present

- Partnered with NTU's Office of International Affairs to become the officially recommended platform for exchange students, linked directly from NTU's exchange program website; achieved 2,500 peak DAU within 2 days of launch (3.56K visitors, 8.16K page views)
- Built interactive map, real-time community features with Next.js 15, TypeScript, and Supabase
- Automated data pipeline aggregating 289 universities and 400+ reviews via scheduled GitHub Actions

AI Course Project, Slide Retrieval RAG Pipeline Mar 2026 – Apr 2026

- Ranked 6th/120 in class Kaggle competition on 200-question retrieval over a 215-page image-heavy PDF, using a 3-stage visual RAG pipeline (ColQwen2.5 retrieval, Visual HyDE, VLM reranking)
- Iterated from 0.460 to **0.933** accuracy through systematic ablation across 16 phases

Awards and Competitions

Semi-finalist, Taipei CodeFest Hackathon 2025 (City Run) Nov 2025

Champion, 2024 International ICT Innovative Services Awards (EventMate) Sep–Oct 2024

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: Next.js, React, Vue.js, Node.js, Django, FastAPI, Tailwind CSS, PyTorch, Hugging Face

Databases: PostgreSQL, MySQL, MongoDB, Supabase, Firebase, Redis, DynamoDB, ChromaDB

DevOps & Tools: Git, Docker, Jenkins (CI/CD), Linux, GCP, AWS Lambda